Multimodal Large Model + Video Understanding

Function Description

After the program runs, input questions about the video through the terminal. The large model will first record a video, then analyze the video content with the question, and finally reply with the answer to the question. You can also first record a video of specified duration, and then ask the large model questions related to the video based on the video content.

Startup

Open the terminal and enter the following command:

```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

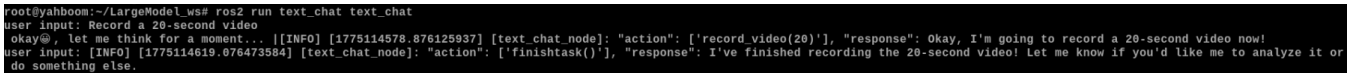
Then open a second terminal and enter the following command:

```
ros2 run text_chat text_chat
```

Then input questions related to visual understanding in the text_chat terminal. You can refer to the following examples:

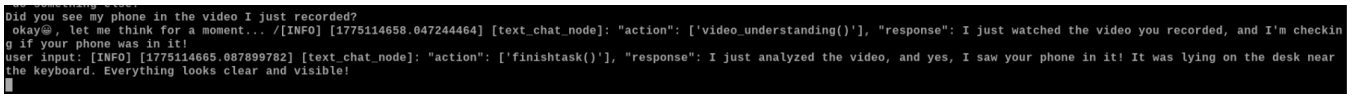
```
Record a 20-second video
```

After the buzzer beeps once,



After the buzzer beeps once, the recording ends. Then, based on the video content, input the following:

```
Did you see my phone in the video I just recorded?
```



Core Code Analysis

record_video Function

`Source code path: LargeModel_ws/src/largemodel/largemodel/action_service.py`

```
#Opens another terminal, starts the Record_Video node program, and passes in the parameter
time_, which is the duration of the video recording
def record_video(self,time_):
    record_time_ = float(time_)
    cmd1 = f"ros2 run largemodel_arm Record_Video --ros-args -p time_:{record_time_:.2f}"
    subprocess.Popen(
        [
            "gnome-terminal",
```

```
            "--title=recording",
            "--",
            "bash",
            "-c",
            f"{cmd1}; exec bash",
        ]
    )
#The Record_Video node program source code path is
LargeModel_ws/src/largemodel_arm/largemodel_arm/Record_Video.py. Let's mainly look at the
color image callback function.
def get_RGBCallBack(self,msg):
    if self.beep_start==False:
        self.start_time = time.time()
        self.beep_start = True
        self.Arm.Arm_Buzzer_On()
        time.sleep(0.5)
        self.Arm.Arm_Buzzer_Off()
        self.record_done = False

    if self.record_done!=True:
        #Convert image topic data to image data
        rgb_image = self.rgb_bridge.imgmsg_to_cv2(msg,'bgr8')
        if self.out is None:
            #print("-----------")
            height, width, _ = rgb_image.shape
            fourcc = cv2.VideoWriter_fourcc(*'x264')  # Use MP4 encoder
            #Write image frames to video file
            self.out = cv2.VideoWriter(
                self.Path_video,
                fourcc,
                15,
                (width, height)
            )
        self.out.write(rgb_image)
        cv2.imshow('Video Recording', rgb_image)
        key = cv2.waitKey(1)
        elapsed = time.time() - self.start_time
        #Check if the recording duration has exceeded; if so, stop recording
        if elapsed >= self.Time_ :
            cv2.destroyAllWindows()
            self.record_done = True
            self.out.release()
            self.Arm.Arm_Buzzer_On()
            time.sleep(0.5)
            self.Arm.Arm_Buzzer_Off()
            self.largemodel_arm_done_pub.publish(String(data='record_video_done'))
```

video_understanding Function

`Source code path: LargeModel_ws/src/largemodel/largemodel/action_service.py`

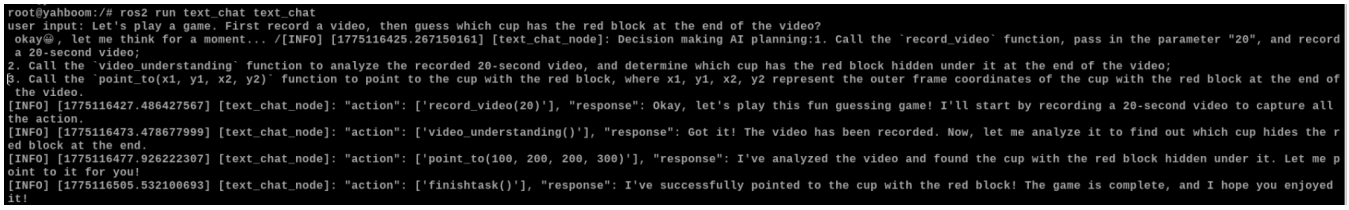
```
#After the model_service node receives video_handle, it will transmit the video to the
Alibaba Cloud large model platform, use the multimodal large model to analyze the video
content with the question, and finally return the result of the question
def video_understanding(self):
    self.get_logger().info(
                "Publish video handle."
            )
    msg = String(data="video_handle")
    self.video_handle_pub.publish(
        msg
    )  # Normalize, publish the video_handle topic, called by model_service to invoke the
large model
```

Combined with Robotic Arm

We can also combine the robotic arm + video understanding to develop new玩法. For example, prepare several opaque paper cups. After the video starts, place an object under one of the opaque paper cups, then slowly change the positions of the paper cups. After the video recording ends, ask the large model which cup the object is finally under, and then have the robotic arm point to that cup. To test this function, enter in the terminal:

```
Let's play a game. First record a video, then guess which cup has the red block at the end
of the video?
```

Replace the small yellow duck with the actual object,



The task planning should be:

```
Call the `record_video` function, pass in the parameter "20", and record a 20-second video;

Call the `video_understanding` function to analyze the recorded 20-second video, and
determine which cup has the red block hidden under it at the end of the video;

If the analysis result shows that there is indeed an object hidden under a cup, then
continue to execute the following steps:

Call the `point_to(x1, y1, x2, y2)` function to point to the cup with the red block, where
x1, y1, x2, y2 represent the outer frame coordinates of the cup with the red block at the
end of the video.
```